**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 071 · Lesson 071 · Pipelining and Computational Methods Review

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What is a heuristic?

Q2. When does backtracking undo a choice?

Q3. What does throughput measure?

Q4. Rising-edge D flip-flop Q=1, D=0 immediately before active edge. What is Q after the edge?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

pipeline — Overlapping multiple processing stages.

throughput — Work completed per time.

latency — Time for one task to finish.

stall — Delay from dependency or resource issue.

## New learning

Observe — Pipelining overlaps separate stages of different tasks; it increases throughput after filling the pipeline, not necessarily single-task latency.

Explain — Four tasks on a three-stage pipeline complete in six ideal cycles, versus twelve when each task runs all stages in isolation.

Evaluate — Choose the computational method that fits: decomposition for parts; backtracking for reversible choices; heuristic for guided estimates; visualisation for patterns.

## Worked example

Find ideal cycles for 3 tasks through 2 single-cycle stages.

1. Explain — Start a new task each cycle while earlier tasks advance; last task finishes cycle4.

# We Do · Practise with guidance

For 5 tasks with 3 equal-length pipeline stages, predict ideal cycles.

```
3+5−1=7 cycles; sequential total=15 cycles.
```

Explain

This is a throughput model assuming one task enters each cycle and no stall.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Draw ideal 2-stage pipeline schedule for 3 tasks and state completion cycle. [6]

### Question 2

For 6 tasks and 4 one-cycle stages, compare pipelined and sequential ideal times. [6]

### Question 3

Match backtracking, data mining, visualisation and pipelining to four problems: maze with dead ends, shopping patterns, charting results and overlapping video processing. Explain each. [8]

# Exit Ticket · Answer before checking

A pipeline always divides each individual task's latency by its number of stages.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Estimate guiding choices, not necessarily guaranteeing optimum.

Q2. When the branch cannot produce a valid result.

Q3. Completed items per unit time.

Q4. Q becomes 0; D is sampled at the active clock edge.

### You Do pages · Independent questions

Q1. T1 occupies stages at cycles1,2 [2]; T2 at cycles2,3 [1]; T3 at cycles3,4 [1]; finishes cycle4 [2].

Q2. Pipeline 4+6−1=9 cycles [3]; sequential 4×6=24 [2]; assumes no stall [1].

Q3. Backtracking maze [2], data mining shopping [2], visualisation charts [2], pipelining overlapping stages [2].

### Exit Ticket · Model answer

False. Overlap typically improves throughput.

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**